

VEINS: Engineering Playbook for Self-Healing Autonomous Fleets

1. Executive Framing: The Convergence of Software and Hardware Resilience

In the design of high-stakes autonomous systems, we must mandate a strategic departure from "rigid" hardware—static structures waiting for failure—toward "malleable" cyber-physical organisms. Traditional human-driven maintenance is a catastrophic bottleneck in autonomous fleet operations; the latency between structural trauma and human intervention represents a systemic vulnerability. To achieve true resilience, we must deploy "Immortal Systems" that autonomously orchestrate continuous self-repair. The VEINS (Vaporization of Sacrificial Components) process serves as the foundational circulatory system for this paradigm, allowing hardware to mimic biological organisms by "bleeding" healing agents into structural fractures in real-time.

Architectural Shift: Traditional vs. Immortal Systems

Metric, Traditional Systems, Immortal Systems

Control Flow, "Human-driven (Localized "Tennis Match")", Agent-driven (Continuous Autonomous Loop)

Scale of Control, Monolithic (Single Point of Failure), Multi-Agent Consortium (Specialized Nodes)

Failure Response, External Reconstruction/Replacement, Autonomous Self-Healing/Vascular Repair

Logic Type, Sequential/Deterministic Code, Neurosymbolic (AI Reasoning + Pure Functions)

Resilience, Brittle (Rigid Components), Malleable (Vascularized Composites)

This transformation begins with the fabrication of the system's "nervous system"—the physical sensors and vascular networks required to feel and respond to trauma.

2. Phase I: The Beginning — Physical Perception and Microvascular Fabrication

The strategic foundation of an immortal fleet lies in high-fidelity perception at the edge. The composite material must function as a sensitive skin, utilizing the VSC process to transform structural panels into diagnostic sensors that detect micro-fractures before they escalate into catastrophic airframe failure.

Fabrication Specification: The VSC Process

The **Vaporization of Sacrificial Components (VSC)** process creates a 3D microvascular network within solid aerospace composites without compromising structural integrity.

- **Methodology:** 300-micron PLA (Polylactic Acid) monofilaments, infused with tin-based catalysts (tin oxalate), are stitched into woven fabric reinforcement.
- **Encapsulation:** The assembly undergoes **Vacuum Assisted Resin Transfer Molding (VRTM)**, encapsulating the fibers in a high-strength structural epoxy matrix.
- **Thermal Post-Cure:** The composite is heated to **150°C**. This activates the catalyst, triggering rapid depolymerization of the PLA into lactide gas.

- **The Physics of Diffusion:** At 150°C, the epoxy matrix becomes "mathematically permeable." This allows lactide gas to diffuse through the matrix and off-gas into the vacuum without causing internal pressure spikes or structural rupture.
- **Verification:** Vascular continuity is verified using **eGaln (Eutectic Gallium Indium)** . This liquid metal is radio-opaque, allowing for X-ray microtomography to confirm a perfect, interconnected 3D circulatory system free of blockages.

Nessie Edge Neurosymbolic Parsing

To isolate the "signal" of a structural crack within the acoustic "noise" of a jet engine, the system employs a three-tier hierarchical parsing framework that structures raw data into a **Causal Graph** :

1. **Symbolic Cache:** Matches acoustic signatures against a database of known fiber breakage.
2. **Retrieval-Assisted Semantic Matching:** Compares anomalous vibrations against a historical vector space of stress events.
3. **Localized SLM Logs:** A Small Language Model (SLM) at the sensor level interprets novel vibrations, converting them into structured event templates (e.g., "Severity Level 4 stress fracture detected at XYZ coordinates").

Reflective Monitoring: The Vigil EMO-bank

While Nessie detects hard fractures, the **Vigil EMO-bank** identifies "soft failures"—logical brittleness where the software is functioning but struggling. Raw server logs are converted into "effective states" (Anxiety, Frustration, Relief) using a **0.5-half-life decay function** . This captures "emotional" aggregates that signal systemic fragility, such as a material's increased flow resistance or API latency, before a physical failure occurs. Physical sensation must now be linked to a centralized fleet orchestration layer for strategic action.

3. Phase II: The Middle — Fleet Orchestration and Deterministic Reasoning

Orchestration is bifurcated between long-term ground-based strategy and localized real-time reflexes. This solves the "latency-vs-reasoning" trade-off: an aircraft cannot wait for a cloud-based inference while a crack propagates at the speed of sound.

The Multi-Agent Consortium

The **Non-Real-Time RIC (RAN Intelligent Controller)** architecture utilizes a ground-based reasoning consortium. By employing heterogeneous models (**GPT, Gemini, Anthropic**) simultaneously, the system mitigates model bias. These agents convert all raw data/HTML into **Markdown** to optimize the **Token Economy** and prevent **Attention Mechanism Degradation** . By stripping boilerplate layout code, the models focus strictly on the factual signal rather than being "mathematically distracted" by metadata.

Deterministic GitHub Integration: The MCP Prohibition

To manage "Digital DNA" and code patches, the use of **Model Context Protocol (MCP)** is **strictly forbidden** .

- **The Risk:** MCP dynamically injects the schemas of 600+ tools into the AI's context window. This "universal remote" approach causes **attention dilution**, where the model's self-attention weighs irrelevant parameters, leading to "flickering failures" and non-reproducible errors.
- **The Mandate:** Use hard-coded **Python functions** for GitHub commits. This treats integration like a "light switch" rather than a "smart hub," ensuring semantic reasoning is handled by the AI while execution remains deterministic and side-effect controlled.

Tactical Near-Real-Time Reflexes

For physical repair execution, the aircraft utilizes **FPGAs (Field Programmable Gate Arrays)**. These are physically rewired at the logic-gate level to serve as a **"Spinal Reflex."** Unlike sequential CPUs, the FPGA triggers repair algorithms in sub-milliseconds, bypassing software instruction cycles to arrest crack propagation instantly. This autonomic reflex provides the signal for the physical execution of the vascular repair.

4. Phase III: The End — Physical Execution and Vascular Restoration

Single-use chemistry is the primary "Thorn" in self-healing research. For a fleet to be "Immortal," it must survive multi-event trauma and prevent internal system clogs.

State-Forked Dual-Vascular Networks

To prevent **"Fatal Thrombosis"** (premature polymerization inside the veins due to vibration or temperature), the system deploys two parallel, interlaced capillary networks:

- **Reservoir A:** Raw liquid resin.
- **Reservoir B:** Chemical hardener.
- **The "So What?":** Isolating components ensures the fluids remain liquid for years. Premixed epoxy clogs its own arteries, committing "structural suicide" before a crack even forms. These components only meet within the site of trauma.

Vellum-Style Recursive Mixing

The repair loop follows **Vellum's recursive control flow**, utilizing the crack void as a mixing chamber. The sequence is as follows:

1. **Detect:** Nessie identifies coordinates and volume.
2. **Calculate:** The FPGA determines the required pulse volume.
3. **Pulse:** **Nordson EFD pneumatic pumps** inject measured volumes of Resin A and Hardener B.
4. **Mix in Void:** Localized trauma provides the chamber for polymerization.
5. **Reverse Pressure:** A pneumatic vacuum sequence is immediately executed to clear the terminal ends.

Arterial Maintenance

To ensure the system remains viable for a decade, the pneumatic vacuum-clearing sequence is mandatory. By reversing pressure, unmixed fluids are pulled back into their respective

reservoirs. This prevents clogs at the vascular terminals, preserving the network for future repair events.

5. Governance Layer: The FRG Triplet and Safety Thresholds

The primary risk in autonomous fleets is **Metastability**, where the cure becomes worse than the disease—such as a repair rerouting traffic so aggressively it crashes secondary clusters.

The FRG Triplet

Every healing action must be bounded by a mathematical governance break to ensure "bounded time convergence":

- **Failure Model:** Definition of survived corruptions, specifically including **Byzantine Failures** (where a node or sensor actively lies about its state).
- **Repair Strategy:** The mechanism of correction (e.g., active redundancy or cryptographic self-stabilization).
- **Guarantee:** The promise of return to a safe state (e.g., consistency within 50ms).

RBT Diagnostic Protocols and the Affinity Gap

Vigil utilizes **Roses, Buds, and Thorns (RBT)** to categorize systemic health. This is calculated via the **Affinity Gap**—an unsupervised machine learning mechanism that maps metrics into a **high-dimensional vector space**. The "Gap" measures the mathematical distance between healthy clusters and anomalous pathogens.

- **Roses:** Verified optimal pathways (Intensity > 0.5).
- **Buds:** Emerging efficiencies (Intensity > 0.3).
- **Thorns:** Systematic failures/Anxiety aggregates (Intensity > 0.4).

Case Study: Robin A "Toast Emission" Bug

The **Robin A** failure demonstrates the necessity of this layer. Traditional logs showed zero errors, but Vigil detected "Thorns" (high-intensity anxiety). The agent was emitting "Toast" (notifications) before receiving a "200 OK" database receipt—a **receipt-independent failure**. The solution was a **"TZ receipt strategy"** and **UTC-aware scheduling**, forcing the agent to block notifications until the backend verified the save. This demonstrates the power of identifying "logical brittleness" that hardware monitors miss.

6. Implementation Summary & Safety Rollbacks

Verify the "Immortal" status of the fleet using the following criteria:

- **VSC Continuity Verification:** Confirm 3D microvascular networks via radio-opaque eGaN and X-ray microtomography.
- **Causal Graph Integrity:** Ensure Nessie Edge filters can distinguish jet engine harmonics from Level 4 fiber breakage.
- **Deterministic Execution:** Verify all GitHub integrations utilize hard-coded Python functions; confirm zero MCP presence.
- **FPGA Reflex Latency:** Confirm logic-gate response time for pneumatic pump triggering is < 1ms.

- **Vascular Clearance:** Test pneumatic reversal to ensure 100% removal of unmixed resin from terminal capillaries.
- **FRG Boundary Test:** Simulate a Byzantine Failure (lying node) to verify the Governance Layer blocks metastable repairs.
- **Recursive Self-Design:** Confirm the reasoning agent can identify inefficiencies in the repair loop and propose structural modifications to its own "Digital DNA." By synthesizing emotional diagnostics with vascular hardware, the fleet moves beyond simple functionality toward a persistent machine life that heals, reflects, and survives.